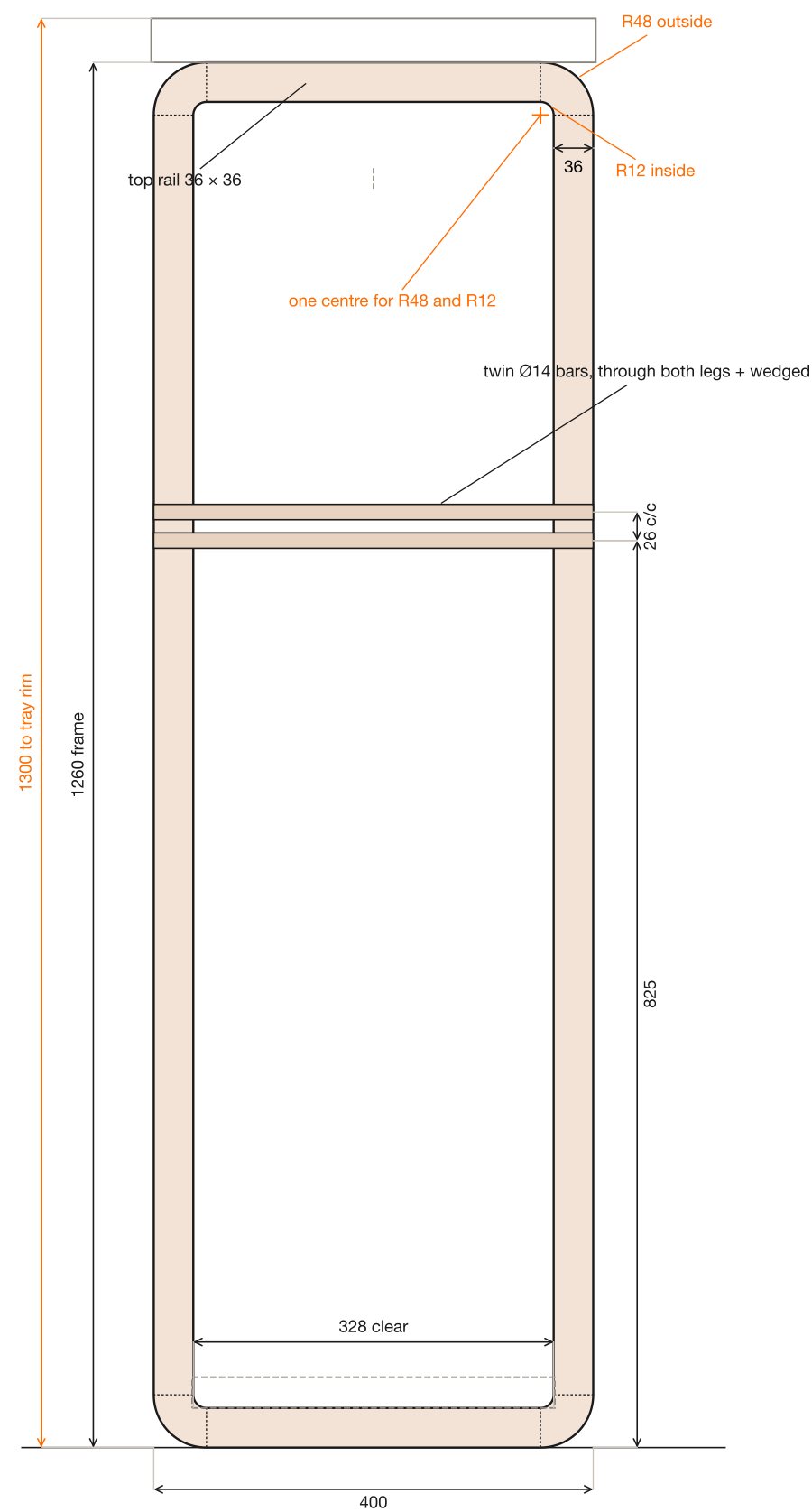
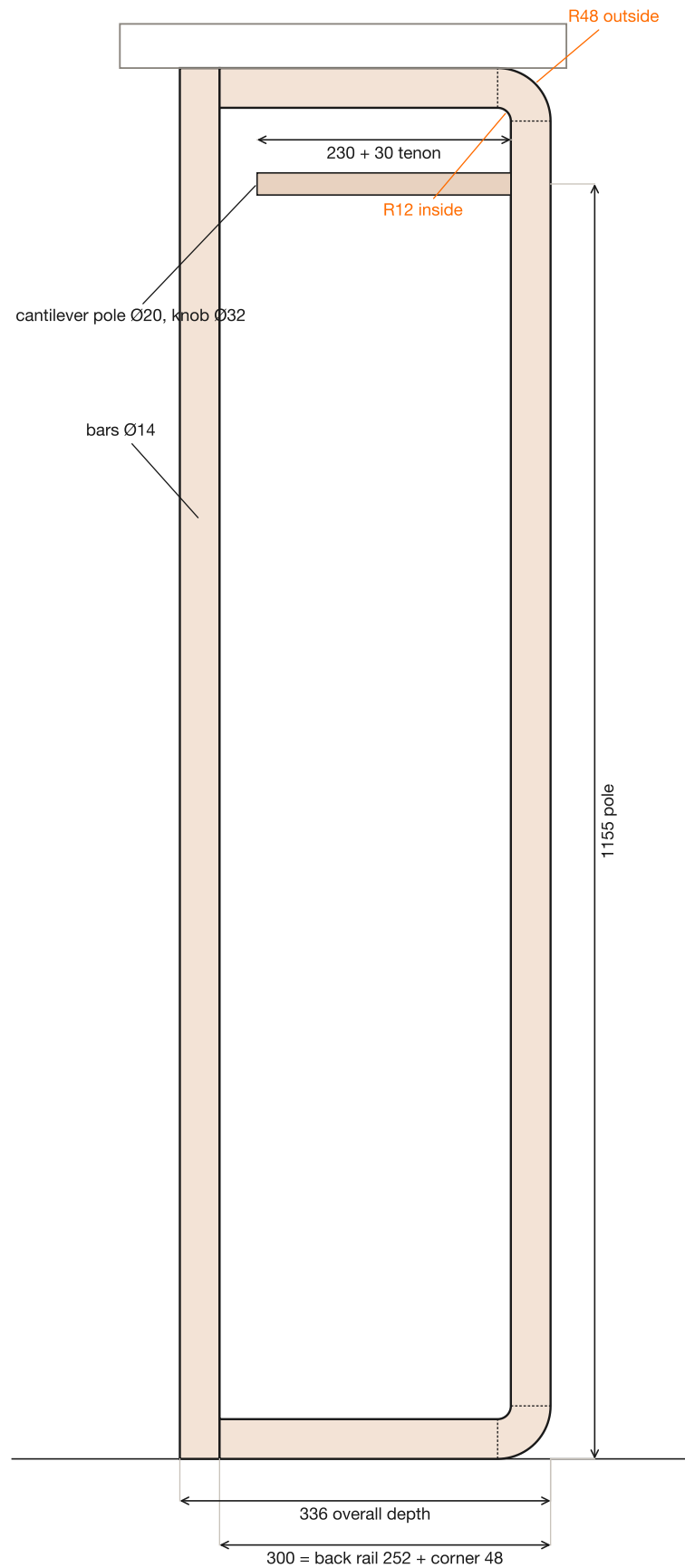


Valet stand — frame

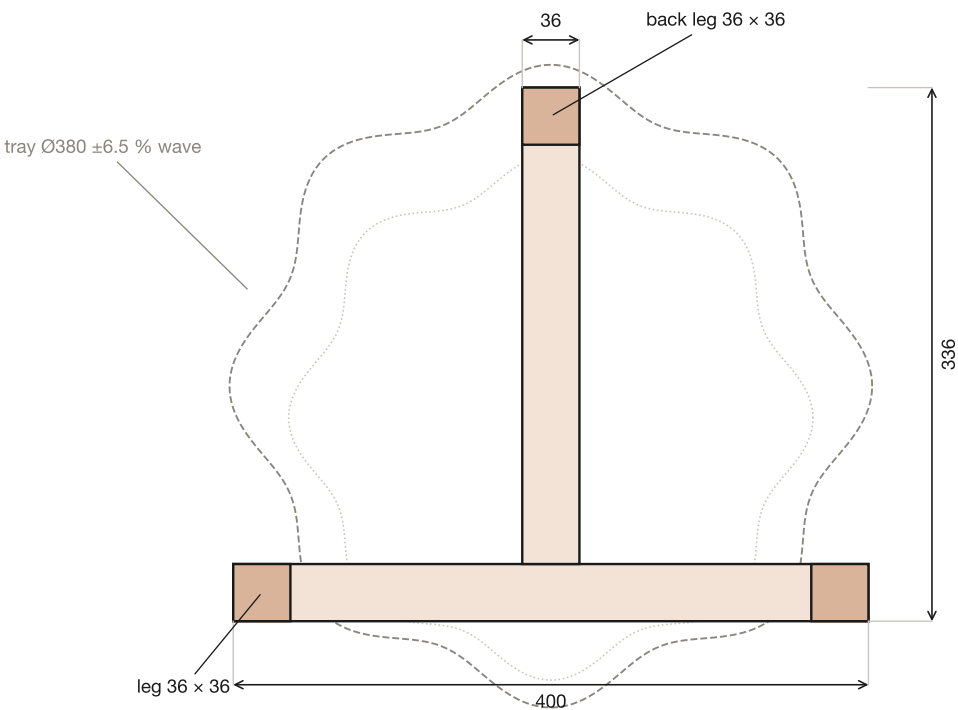
Front elevation · side elevation · plan · tray top 1300 mm · legs + rails 36 × 36 · corners R48 / R12, one centre · all dimensions in mm



FRONT ELEVATION



SIDE ELEVATION



PLAN (tray dashed)

- Frame stock: mahogany (sipo). Every member 36 × 36: legs, rails and corners.
- Corners are concentric (R48 / R12, one centre), so the band stays 36 wide through the bend.
 - Each corner is its own piece, grain along its chord (sheet 3). Joints = dashed lines, square cuts.
 - Every joint: 2× Ø8 × 60 dowels, 35 into the straight part, 25 into the corner.
 - Back rails into the back face of the front rails: 2× Ø8 × 60, 35 + 25.
 - All arrises R4 round-over, routed after glue-up so it runs unbroken over every joint.
 - Twin trouser bars stacked (as drawn): side by side would leave 4 mm of wall.

Cut list · order of work · tray top 1300 mm · legs + rails 36 × 36 · corners R48 / R12, one centre · all dimensions in mm

	PART	QTY	LENGTH	SECTION	WOOD	NOTES
A	Front leg	2	1164	36 × 36	Mahogany	straight, square ends; 2× Ø8 holes 36 deep in each end
D	Back leg	1	1164	36 × 36	Mahogany	as A
B	Front rails (top + bottom)	2	304	36 × 36	Mahogany	straight, square ends; 2× Ø8 holes 36 deep in each end
E	Back rails (top + bottom)	2	252	36 × 36	Mahogany	straight; front end into the back face of B, back end into a corner piece
K	Corner pieces	6	blank 80	43 × 36	Mahogany	grain along the chord; R48 / R12 one centre; finished 39.5 × 67.9; 2× Ø8 holes 26 deep each end (sheet 3)
G	Trouser bars	2	400	Ø14	Mahogany	through both front legs at 825 / 851 from floor = 777 / 803 from leg bottom
H	Hanger pole	1	260	Ø20	Mahogany	30 mm into D at 1155 from floor = 1107 from leg bottom; knob Ø32 × 24
—	Joint dowels	28	60	Ø8	beech	14 joints × 2 (6 corners × 2 ends + 2 back rails); 35 + 25

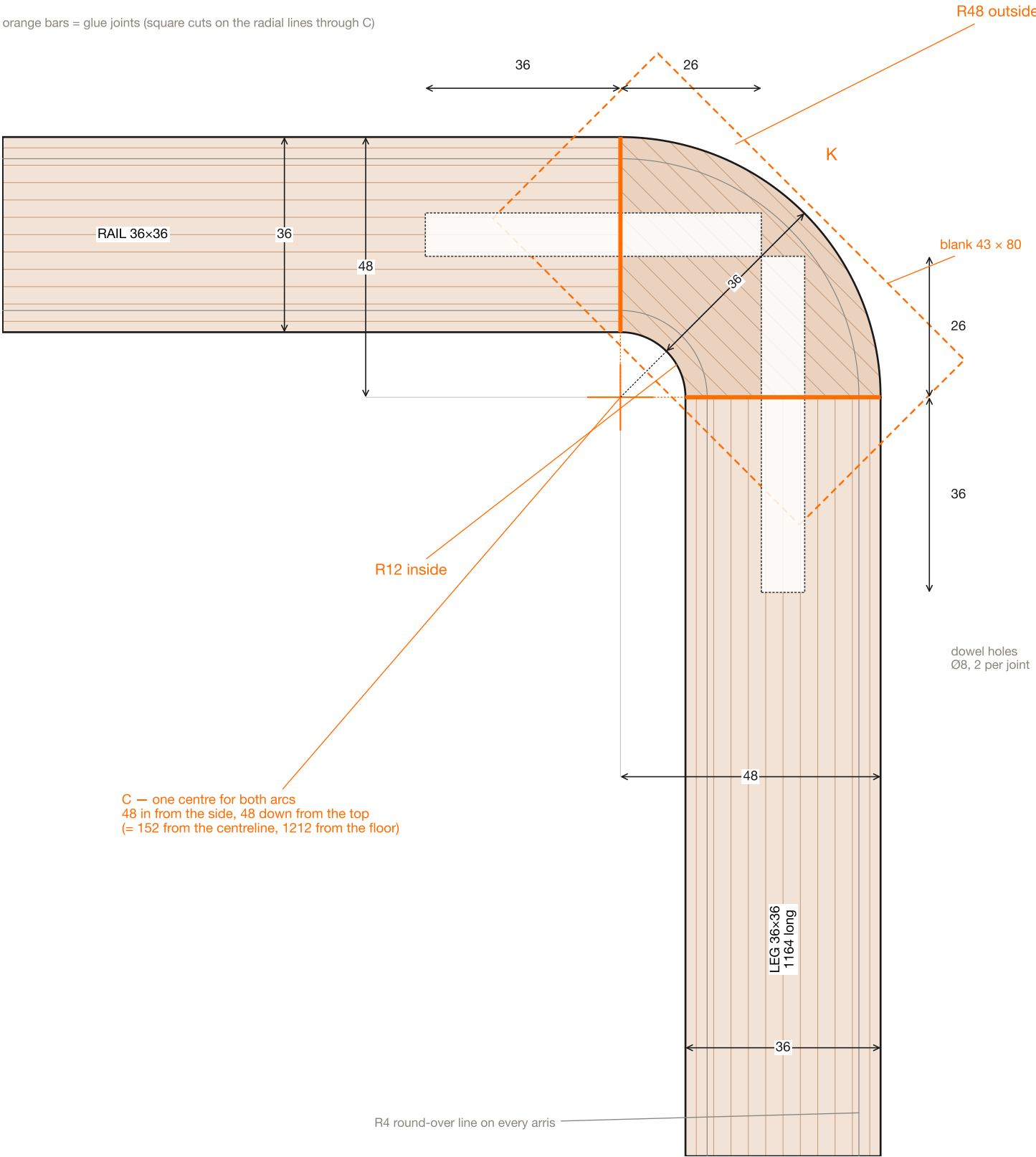
Corner pieces: see sheet 3 (geometry, maths, blank layout, 1:1 template).

Order of work

1. Mill all mahogany to 36 mm; rip legs to 36 × 36 and cut them to 1164.
2. Cut the straight rails: 2× 304 (front) and 2× 252 (back), 36 × 36.
3. Rip one strip 43 × 36, ~520 long; crosscut 6 blanks 80; bandsaw + flush-trim to the 1:1 template on sheet 3.
4. Dowel holes Ø8: 36 deep in legs/rails, 26 deep in corners and the back face of B; bar holes through both front legs.
5. Glue the front ring (2 legs + 2 rails + 4 corners + bars), then the back C (back leg + 2 rails + 2 corners), then join.
6. Drill the Ø20 pole hole into D at 1155 (1107 from its bottom), 30 deep; rout R4 on every arris; finish.

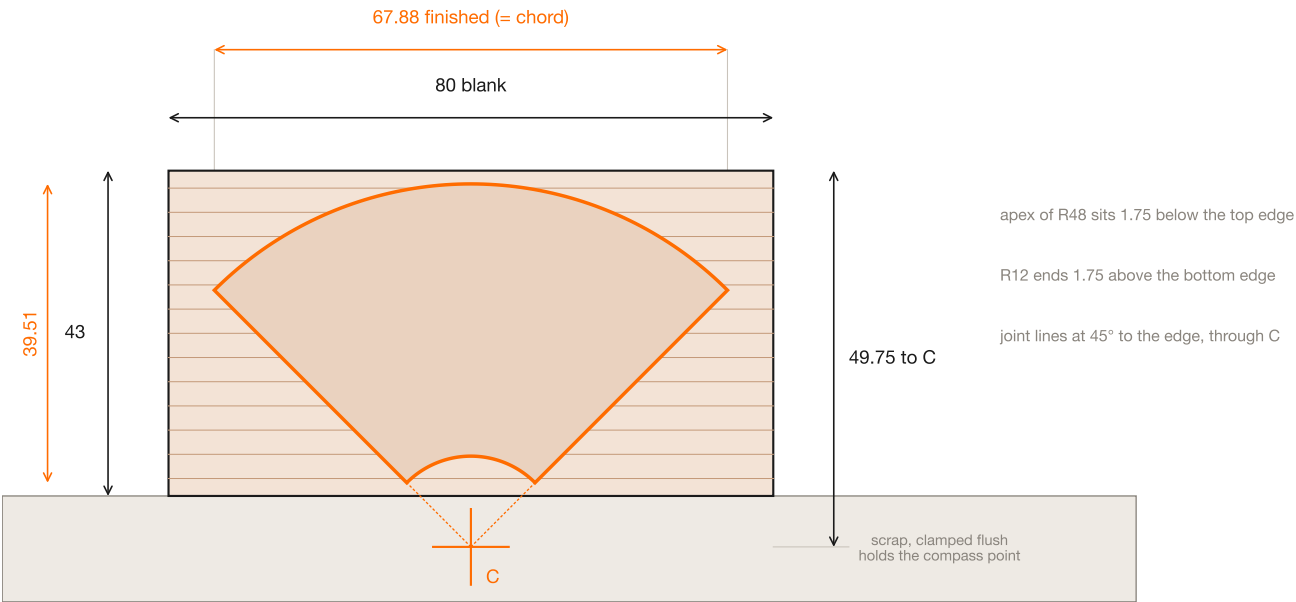
A · FRONT VIEW, TOP-RIGHT CORNER, 1:1

orange bars = glue joints (square cuts on the radial lines through C)

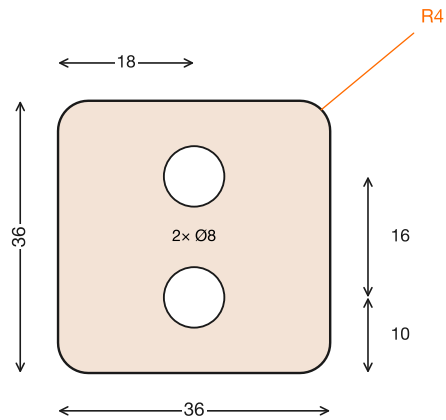


B · THE BLANK, 1:1 — GRAIN RUNS ALONG THE CHORD

cut out the orange shape and glue it to 6 mm MDF = your template



C · END OF ANY MEMBER, 1:1



THE MATHS

Band $t = 36$ (legs = rails = corners)
Outside $R_o = 48$ · inside $R_i = R_o - t = 12$
Same centre C, so the band is 36 wide all round
 $C = (152, 1212)$ from centreline / floor
Leg = $1260 - 2 \cdot 48 = 1164$
Front rail = $400 - 2 \cdot 48 = 304$
Back rail = $300 - 48 = 252$
Blank length = chord = $2 \cdot 48 \cdot \sin 45^\circ = 67.88$
Blank width = $48 - 12 \cdot \cos 45^\circ = 39.51$
 $+ 2 \cdot 1.75 = \text{rip } 43$ · $+ 2 \cdot 6 = \text{cut } 80$
Arc lengths: out 75.4 · centre 47.1 · in 18.8
Grain at the ends crosses the band at 45° :
fibres $36 / \sin 45^\circ = 50.9$ long — no short grain
Dowel into corner: centre 30 from C, edge 34;
3 mm skin under R48: depth $\leq \sqrt{(45^2 - 34^2)} = 29.5$, use 25
Pole Ø20 at 1155: top 1165, leg-top holes
end at 1176: 11 mm clear

MAKING THE 6 CORNERS

1. Rip one strip 43×36 , ~520 long, from the leg stock.
2. Crosscut 6 blanks 80 (same strip = same colour).
3. Template from B on 6 mm MDF, or mark direct:
C at 49.75 from the top edge, mid-length;
compass R48 and R12; joint lines 45° through C.
4. Bandsaw 1–2 mm proud; flush-trim to the template.
5. Ends: sand to the line on a disc + 45° fence, square.
6. Ø8 holes 26 deep each end, 18 in from the outside, 10 and 26 from the front (C).
7. Glue up, then R4 round-over on all arrises.
Inner R12 > bearing radius 6.35 — bit follows it.



Scale check: this bar must measure exactly 50 mm on paper. If not, print again at 100 % / "actual size".